

Fall-Prevention Exoskeleton: Undergraduate Student Researcher

Rowan University · Jan 2024 – Present

Falls are a leading cause of injury for the elderly and people with limited mobility. The Fall-Prevention Exoskeleton is a pneumatically actuated, wearable device that detects perturbations in the wearer's gait in real time and corrects them before a slip or fall occurs, an assistive layer of safety for those who struggle with walking, supporting confidence and independence. Over the most recent semester the team completed a major hip-to-knee chassis redesign, a ground-up rebuild of the sensing system and test bench, and an instrumented pneumatic bench-test campaign. My ownership covers the sensing, test-bench, force-characterization, and pneumatic-hardware subsystems detailed below.

- **Engineered a wearable 8-node motion-capture system:** built over a full-duplex RS-485/UART transceiver matrix using shielded flexible cables to eliminate signal noise during human trials.
- **Routed custom Teensy-based mainboards:** aggregated high-frequency BNO085 IMU data streams (quaternion, gyroscopic, and accelerometer data at up to 400 Hz per node) to a central Raspberry Pi gateway for real-time predictive gait detection.
- **Executed exoskeleton bench testing:** ran 192 total trials using string potentiometers, pressure transducers, and a single load cell across hip abduction and knee extension setups within a hardware-in-the-loop (HIL) pneumatic test matrix.
- **Researching NMES integration:** currently in the research phase for a neuromuscular electrical stimulation system to assist in motor control, scoping the low-level electronics and multi-channel current-regulation architecture it will require.

IMU System V2: Wearable Sensing Network

An eight-sensor system built on BNO085 inertial measurement units placed at the thighs, calves, ankles, lower back, and upper back, fully contained in custom 3D-printed enclosures with adjustable straps for wearability. The system is built on a custom PCB with 16 full-duplex RS-485 transceivers (TI SN65HVD) and a Teensy 4.1, selected for its 600 MHz clock and eight independent hardware UART channels, exactly matching the sensor count for simultaneous acquisition from all IMUs. Custom shielded Cat6 twisted-pair cables, length-adjustable and modular on both ends, provide EMI resistance directly adjacent to pneumatic solenoid valves and switching regulators.

The previous I²C-based iteration had three fundamental limitations: no differential signaling (EMI susceptibility), cable lengths beyond rated specification, and multiplexer latency compounded by 1–3 ms of sensor-fusion clock stretching per IMU. Modeled end to end, a frame took 2.34 ms of transfer plus 8×2 ms of clock stretching = 18.34 ms, exceeding the 10 ms window required for 100 Hz sampling. I documented this failure analysis, then rebuilt the system around UART over RS-485: industrial-grade noise immunity, cable ratings beyond 100 m, and point-to-point channels on which all eight sensors transmit simultaneously.

Parameter	I ² C (actual)	UART (921,600 bps)	UART (6 MHz max)
Frame latency	18.34 ms	0.33 ms	0.051 ms
Bus utilization at 100 Hz	183.4%	3.3%	0.51%



IMU System V1, the original I²C architecture: fully assembled custom multiplexer PCB, the eight body-segment sensor nodes on their straps, worn testing of the first-generation array, and the belt-mounted acquisition hub with per-segment channels.

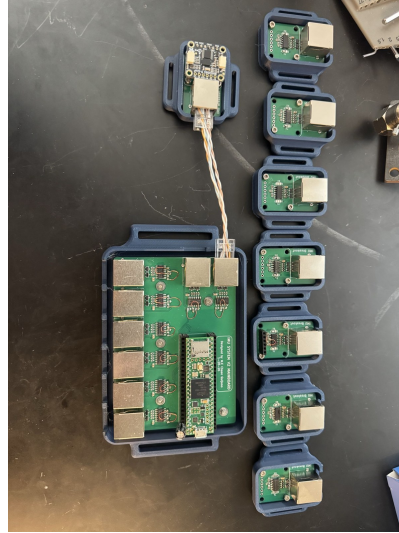
The Teensy 4.1 owns all sensor communication and streams normalized quaternion data over serial to a Raspberry Pi running the gait-detection algorithm; when the algorithm flags abnormal gait indicative of a potential fall, it

issues a serial command to a dedicated control board that actuates the pneumatic cylinders to assist the user. This separation of concerns dedicates the Pi to gait analysis, leaves headroom for algorithm growth, and isolates data acquisition from processing. A 10-minute sustained trial logging all eight IMUs simultaneously validated the system:

Metric	Result
Sampling rate	400 Hz sustained (ref.: Delsys 120 Hz, Xsens 240 Hz)
Packet interval	<2.5 ms consistent
Packet loss	0.0%
Mean orientation error	0.003% (unit-quaternion magnitude deviation)



IMU System V2 worn for data acquisition: each sensor at its designated anatomical location on custom adjustable straps.



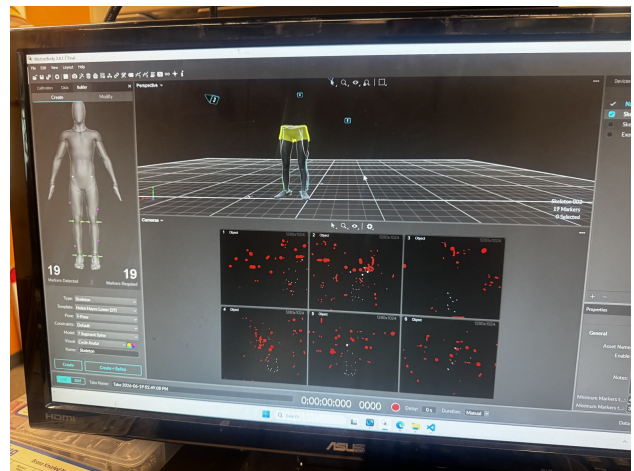
Custom Teensy 4.1 mainboard and BNO085 sensor nodes: simultaneous logging at up to 400 Hz per node over the full-duplex RS-485 matrix.



Assembled 8-node array: compact, low-profile modular housings keep hardware light, secure, and unobtrusive during human movement testing.



Treadmill walking trial with the IMU system worn, validated simultaneously against an OptiTrack optical motion-capture system, with reflective markers tracked alongside the IMU array.



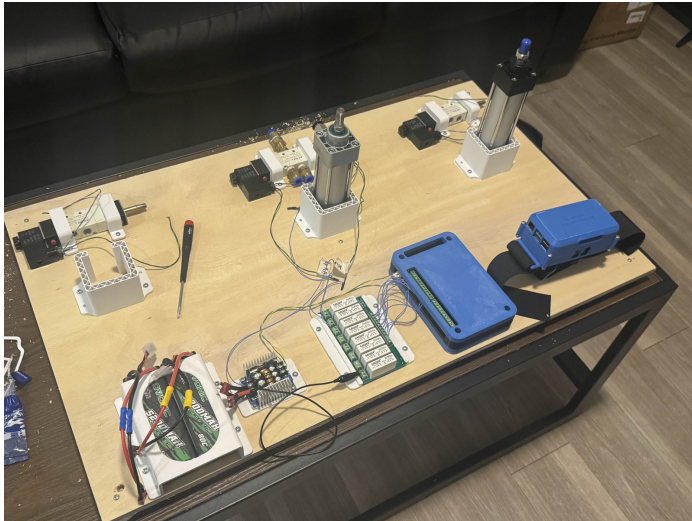
OptiTrack Motive software reconstructing the reflective-marker skeleton in real time during the walking trials, serving as optical ground truth alongside the IMU data.

Test Bench V2

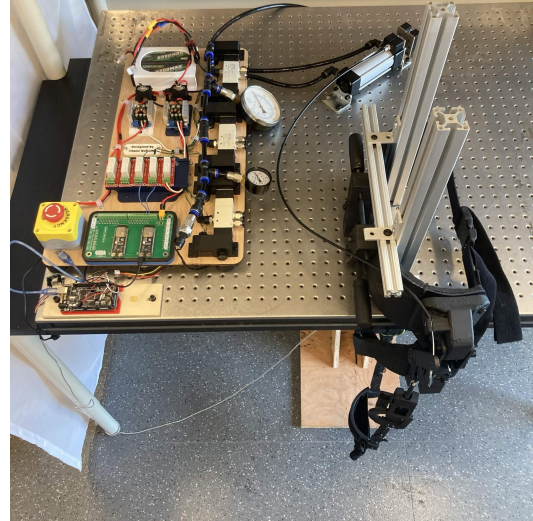
The original plywood test bench proved the concept but had an impractical footprint, and its 3D-printed cylinder mounts lacked the rigidity required for reliable actuation under pneumatic load. The revised bench is built on a

laser-cut wood chassis that significantly reduces the footprint while improving structural rigidity, and adds the functional features testing actually needed:

- Emergency-stop button for immediate system shutoff during testing.
- Dedicated toggle switches for the 15 V and 5 V power rails for manual line-level control during debugging and evaluation.
- A single centralized pneumatic line feeding all solenoids, for simplified routing and fast pressurization between test cycles.
- An integrated pressure transducer for real-time pressure readings during data acquisition, and improved cable management to reduce wiring clutter and accidental disconnections.



Original plywood test bench: 3D-printed component mounts, series LiPo power, and the first pneumatic cylinder and solenoid layout.

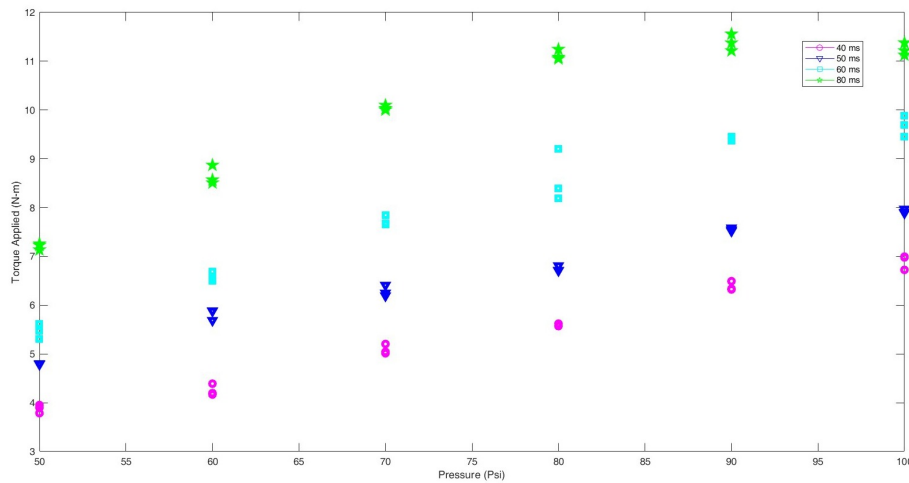


Test Bench V2 laid out and prepared for hip-abduction load-cell data acquisition: E-stop, switched power rails, centralized pneumatic line, and integrated pressure transduction.

Hip Abduction Load-Cell Testing

Characterizing the exoskeleton's force output was a primary objective. The chassis was fixed to a threaded steel table with a horizontal load-cell fixture mounted parallel to the exoskeleton leg, connected by a rope under tension with the static tension accounted for in all readings. A linear calibration model was derived from the manufacturer's documentation and verified through static trials. A standardized procedure governed every run: regulator setpoint, actuation delay configured in the control-board DAQ firmware, and a MATLAB DAQ script with output files named by trial number, pressure, and delay for traceability.

A total of 96 trials were completed across a matrix of four actuation delays (40, 50, 60, 80 ms) and six supply pressures (50–100 PSI). The collected data has been validated as viable, with trend analysis across the parameter space ongoing.



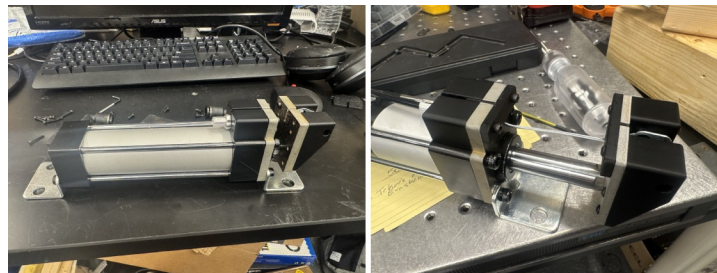
Hip abduction load-cell results: supply pressure (PSI) against applied torque (N·m) across the 96-trial matrix.



Hip mechanism on the force-testing stand with the load cell inline, the fixture behind the trial campaigns.

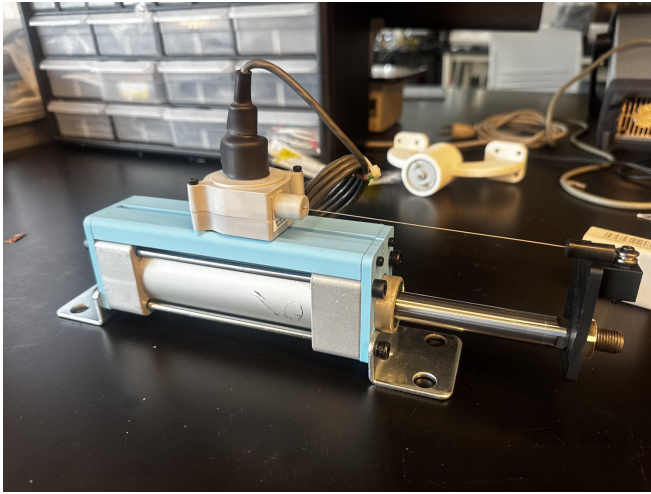
Modular Pneumatic Cylinder Assembly

Fixed, crimped Bowden-cable fittings on the pneumatic cylinders made modifications and component swaps time-consuming in earlier iterations. I addressed this with a modular cable-retention mechanism developed by reverse-engineering a bicycle brake-lever assembly: a keyed steel piece on a swiveling axis locks the cable end-bead in place, allowing tool-free cable removal. The first-generation assembly was constructed from 1/4" waterjet-cut steel and ABS to maintain structural integrity under cyclic loading.

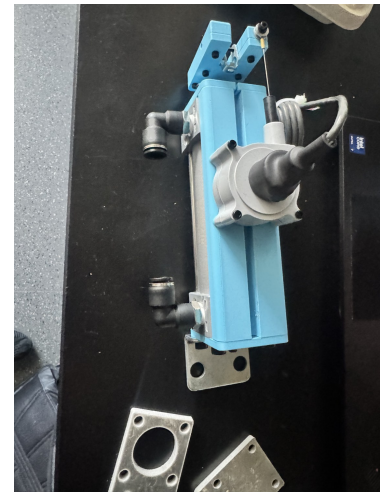


First-generation modular pneumatic cylinder assembly: tool-free Bowden-cable retention adapted from a bicycle brake-lever mechanism.

The second-generation mechanism adds a custom mount for a string potentiometer, enabling closed-loop position control in a compact form factor, and replaces the waterjet-cut steel plates with waterjet-cut aluminum for weight reduction ahead of its integration into the wearable backpack.



Cylinder mechanism V2: string potentiometer on its custom mount with the wire routed to the rod end for closed-loop position feedback.



V2 assembly with waterjet-cut aluminum mounting plates, replacing the previous steel for weight reduction.

Published work from this research program. The exoskeleton program builds on the following publications by our group:

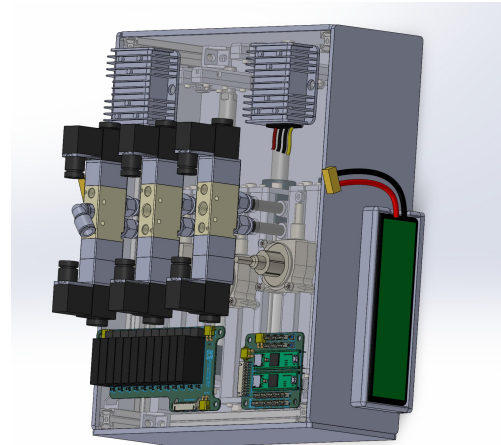
1. V. Varma, S. N. Patel, N. P. Wilson, and M. Trkov, "Characterization of Hip Abduction Exoskeleton for Assistance During Gait Perturbations," *IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, 2024. doi:10.1109/AIM55361.2024.10637061.
2. V. Varma and M. Trkov, "Intersegmental coordination in human slip perturbation responses," *Journal of Biomechanics*, vol. 168, art. 112097, 2024. doi:10.1016/j.jbiomech.2024.112097.
3. V. Varma and M. Trkov, "Investigation of intersegmental coordination patterns in human walking," *Gait & Posture*, vol. 112, pp. 88–94, 2024. doi:10.1016/j.gaitpost.2024.05.010.
4. V. Varma, Z. Roberts, F. Mallick, and M. Trkov, "Estimating Human-Exoskeleton Interaction Forces Using an Instrumented Thigh Brace and OpenSim," *IFAC-PapersOnLine*, vol. 59, no. 30, pp. 353–358, 2025 (5th Conference on Modeling, Estimation and Control, MECC 2025). doi:10.1016/j.ifacol.2025.12.262.

Current Work & Path to Human Trials

Biomechanics testing is now underway, work I carry out alongside PhD student Vaibhavsingh Varma: worn trials with the full system (harness, exoskeleton, and IMU sensing network together) conducted on an overhead safety tether.



Worn biomechanics testing on the overhead safety tether: front, back, and side walking views, and a simulated-fall trial.

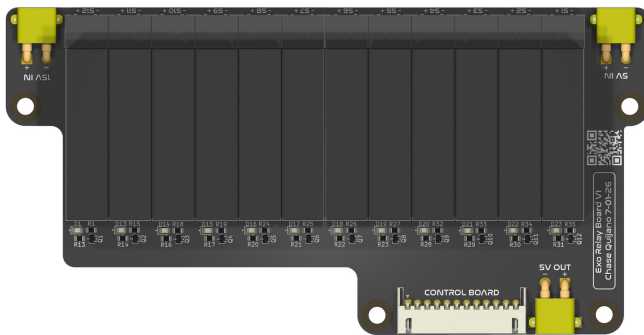


CAD model progress on the exoskeleton backpack design: solenoids, custom relay and control boards, battery, and laser-cut paneling in one enclosure.

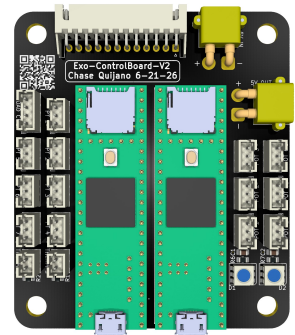
In parallel, I am developing an all-in-one backpack that encompasses the complete control system in a single wearable unit:

- Completely new custom electronics, designed to be as compact as possible: a custom relay board of my design and custom Control Board V2, nearly one-third the size of the previous generation.
- Latest edition of the cylinder mechanism, using string-driven potentiometers for closed-loop control.
- Laser-cut paneling; battery powered, with battery management and monitoring.
- Pneumatics pressurized via replaceable CO₂ canisters for instantaneous release during human trials.

Human trials are projected for this fall semester, with a goal of 20 subjects for testing.



Relay Board V1 render: 12-channel, flyback-protected relay board of my design for the backpack control system, with switched 15 V/5 V distribution and a dedicated control-board header.



Control Board V2 render: primary and secondary microcontrollers, with temperature monitoring across multiple systems plus pressure and voltage monitoring serving as a safety function, and LED status indication.

PhD student Vaibhavsingh Varma and I are preparing for the human trials and collecting the feedback that will shape the final procedure. I have been present for all testing throughout my time on the project, will help run the human trials this fall in the Rowan University biomechanics lab, and will be a co-author on an upcoming publication from this work.